1 A hot air balloon is tied to the ground using a rope. As the wind blows with speed v, the rope makes an angle θ to the horizontal, as shown in Fig 1.1.

For Examiner's Use

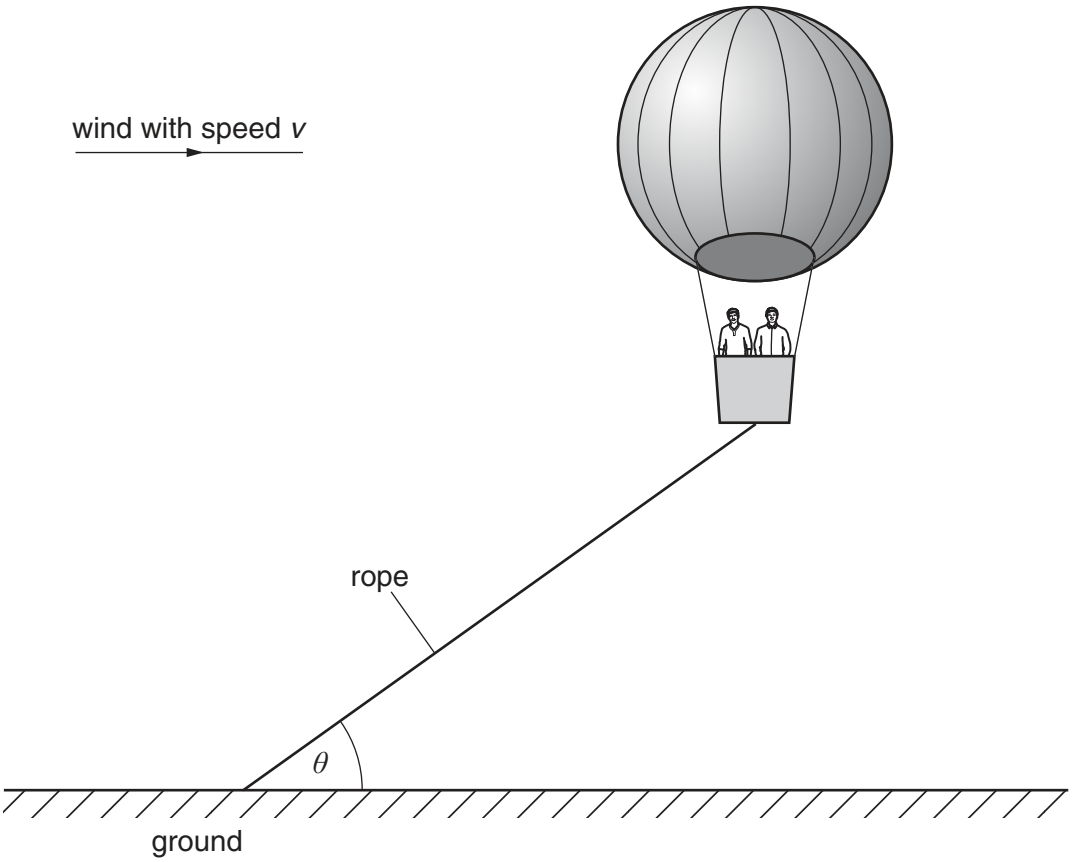

Fig 1.1

It is suggested that $\tan\theta$ is inversely proportional to v^2.

To model the hot air balloon in the laboratory, a balloon filled with helium is used. Design a laboratory experiment using a small helium-filled balloon to test the relationship between θ and v. You should draw a diagram, on page 3, showing the arrangement of your equipment. In your account you should pay particular attention to

(a) the procedure to be followed,

(b) the measurements to be taken,

(c) the control of variables,

(d) the analysis of the data,

(e) the safety precautions to be taken.

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